
ENTROPIC ATTRACTOR NETWORKS: A SCALE-AGNOSTIC, FIRST-PRINCIPLES ROUTE TO INFERENCE AND LEARNING IN BRAINS AND MACHINES

WHITE PAPER, VERSION 0.9.1 (DRAFT)

Tamas Spisak¹

Predictive Neuroscience Lab, Center for Translational Neuro- and Behavioral Sciences
University Medicine Essen, University of Duisburg-Essen

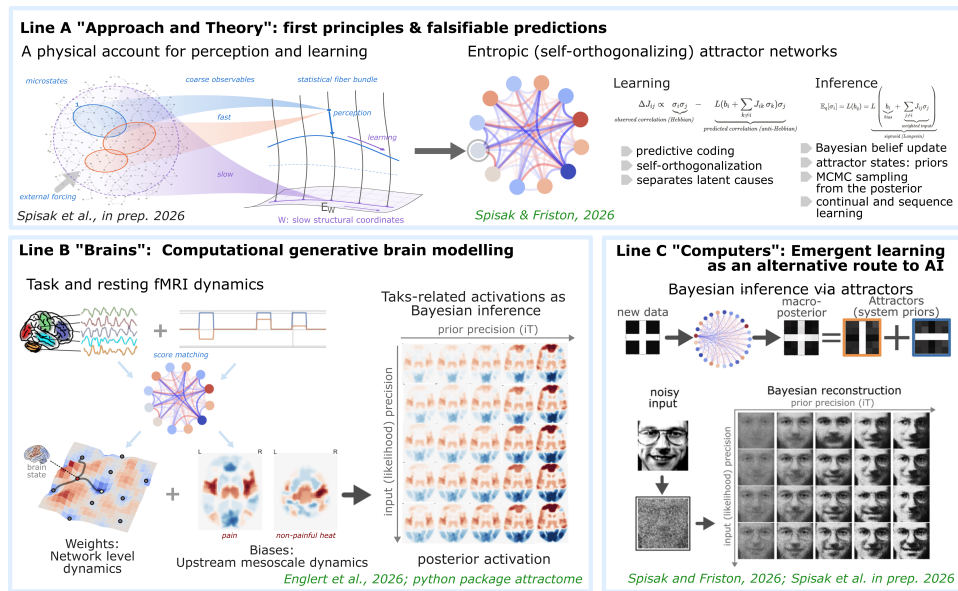
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Contributors: Balint Kincses, Raviteja Kotikalapudi, Robert Englert, Jenya Lerer, Kevin Hoffschlag, Jialin Li, Giuseppe Gallitto (Predictive Neuroscience Lab).

Commenters and advisors (listing does not imply endorsement): none yet. Colleagues whose comments or advice shape future versions will be listed here, with their permission (see About this document).

Abstract

Brains learn continually without overwriting what they know, represent uncertainty and act on internal models of the world, all on about 20 watts. These capacities remain hard for artificial intelligence. Neuroscience, despite a detailed picture of the mechanisms involved, cannot yet explain from first principles why matter organized in this way infers and learns at all. This white paper sets out a research programme that approaches the question from below. Instead of asking what any persisting system must look like, as the free energy principle does, we ask what is minimally required for a piece of matter to infer and learn, and we answer this for a single unit, from a few explicit physical assumptions that can each be checked. In the resulting construction, prior, likelihood and evidence follow from normalization, prediction errors from the geometry of normalized densities, the response function from boundedness, and precision from fluctuations. Perception and learning are fast and slow entropic relaxation. Composed into networks, such units form entropic attractor networks (eANNs), whose network-level properties have already been published: their attractors are learned, approximately orthogonal Bayesian priors, and they learn sequences and resist catastrophic forgetting. Line A develops this theory. Line B fits eANNs to fMRI as generative models of large-scale brain dynamics with testable, mechanistic predictions. Line C develops them as an alternative route to AI, in software and on emerging hardware.



¹Correspondence to: tamas.spisak@uk-essen.de

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Contributors and collaborators: Balint Kincses, Raviteja Kotikalapudi, Robert Englert, Jenya Lerer, Jialin Li, Kevin Hoffschlag, Giuseppe Gallitto (Predictive Neuroscience Lab)

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About this document: its origins, status and how to contribute

This white paper sets down a research vision before it has been fully formalized and tested. It grew out of several years of empirical, simulation and theoretical work, part of it published and much of it still in preparation. Taken together, these results suggest a general and, we think, intuitive picture of how inference and learning can emerge from suitably organized matter, and of how understanding this could help both neuroscience and AI. The purpose of the document is to state that picture explicitly, so that it can be developed rigorously and tested piece by piece.

What follows is a *research programme* in the sense of Lakatos [1978], rather than a finished theory: a small core of assumptions (Section 3), auxiliary hypotheses that can be revised as evidence comes in, and predictions that can fail (Section 7). Its direct value lies in the predictions it makes and the work it organizes, not in the breadth of what it currently explains. We expect some of the ideas below to be wrong, and we have tried to state each one so that its failure would itself be informative.

The document is versioned and will be updated as results arrive. Comments, criticism and collaboration ideas are welcome, in particular from colleagues who work with other approaches, such as biophysical whole-brain models, task-optimized deep networks or the free energy principle. Serious objections will be recorded and answered in later versions, and readers who contribute will be acknowledged by name, with their permission. Contributors are listed on the title page, as are commenters and advisors whose comments or advice shaped a version; being listed records that input and does not imply endorsement of any claim. Responsibility for the content rests with the author.

Executive summary

Brains never stop learning. A new face, a new route home or a new word is added to what we already know without wiping it out, and while this happens the brain keeps track of how certain it is, all on about 20 watts. Artificial intelligence has become very capable, yet these everyday feats remain hard for it. Large models are trained offline, tend to overwrite old knowledge when they learn something new, represent uncertainty poorly and consume a great deal of energy. Neuroscience, for its part, has catalogued many of the mechanisms behind the brain’s abilities, from synaptic plasticity to replay and attractor dynamics. What it cannot yet say is why matter organized in this way should infer and learn at all, and why similar solutions recur in single cells, circuits and whole brains.

We think that both fields are missing the same piece: an account, at the level of physics, of how inference and learning arise in matter. The most ambitious attempt at such an account is the free energy principle (FEP), which holds that any system that persists in a fluctuating environment must look as if it infers the causes of what it senses. Its generality is its strength, but it is also why the principle has been hard to turn into concrete models that can be fitted to data and that could turn out to be wrong.

This white paper describes a research programme that approaches the question from the other end. Rather than asking what any persisting system must look like, we ask what is the least a piece of matter needs in order to infer and learn. We answer this question for the simplest possible building block, a single unit, from a handful of explicit physical assumptions that can each be checked in a given substrate, and only then ask what happens when such units are combined. Several developments make this route feasible now (Section 3.1). The most concrete is a recently published network-level result [Spisak and Friston, 2026]: networks derived top-down from the FEP perform Bayesian inference, store their priors as attractors and learn with

a local rule that drives these attractors towards orthogonality. A bottom-up construction of the unit has to recover exactly these rules, including the timing of the learning signal, without having them built in. It also makes predictions that the network result does not, for example that a unit’s fluctuations equal its gain, and that the shape of its response function depends on its range and resting law (Section 8.1).

In the construction (Section 3), the familiar ingredients of Bayesian brain theories are not postulated but appear as consequences. Normalizing a perturbed physical ensemble turns it into a prior, a likelihood and a model evidence. A prediction error turns out to be the only kind of change that preserves normalization. The unit’s response function follows from its bounded range, and its fluctuations encode the precision of what it represents. Perception becomes fast relaxation, and learning slow relaxation that matches predicted to observed correlations. Composed into networks, such units form what we call *entropic attractor networks* (eANNs). These are stochastic attractor networks whose attractors are learned, approximately orthogonal priors, which learn sequences, and which resist catastrophic forgetting. Because the form of the construction is expected to survive coarse-graining (Section 4.2), models of the same family may describe single neurons as well as whole brain regions, which is what makes functional MRI a legitimate test bed.

The programme tests three hypotheses, each in one line of work:

1. **Matter can infer and learn (Line A, “Approach and theory”).** In their most elementary form, inference and learning are capacities of suitably organized matter rather than algorithms imposed on it. Line A completes the unit-level theory and extends it to hierarchy, action, planning and thermodynamic cost.
2. **Biology exploits this across scales (Line B, “Brains”).** Living systems use this elementary form of inference at many scales and show measurable signatures of it. Line B fits eANNs to fMRI data as generative models of large-scale brain dynamics, tests the theory’s predictions, and asks which computations are carried by large-scale attractors and which by processes below them, using learning, extinction, expectations and pain as test cases.
3. **Machines can do the same, natively (Line C, “Computers”).** Line C develops eANNs as an alternative route to AI, tests them on a neuroscience-inspired benchmark battery, and explores implementations on thermodynamic, memristive, nanomagnetic and other emerging hardware, where noise is a resource rather than a nuisance.

The programme does not promise a full account of the brain’s complexity, but a firmer grasp of its roots: predictions derived from one construction that should hold, in brains and machines alike, at different scales; generative models of brain dynamics whose components have a clear computational meaning; and learning machines in which continual learning and uncertainty are built in rather than added on. Each line can produce valuable results even if the others stall, and every prediction comes with a stated criterion for failure (Section 7).

1. Introduction

1.1 What brains do and machines still struggle with

Modern AI writes, reasons and generates images, but its progress towards biological intelligence is strikingly uneven. Large models are trained offline, on vast data, with global loss functions and with methods shaped by the constraints of von Neumann hardware, the “hardware lottery” [Hooker, 2021]. Four capacities that brains show routinely remain hard for them.

The first is continual learning. In brains, learning and inference are inseparable and never stop, whereas artificial networks trained on new data tend to overwrite what they knew (catastrophic forgetting). The same limitation shows up within a single long task: in a recent comparison, language-model agents given a day of compute fell behind the gains that repeated sampling would predict, whereas expert humans kept improving over two weeks [Mang et al., 2026] (an informal comparison, cited here as motivation rather than evidence). From the perspective developed below, this is less a matter of protecting old weights than of missing timescales: inference runs, but there is no slower level of structure that keeps adapting while the task is performed, and no level slower still that consolidates what was learned (Sections 3, 4.3 and 5). The second is uncertainty: brains act on calibrated uncertainty, while most artificial networks represent it poorly or only through add-on machinery. The third is frugality. Brains learn with local plasticity driven by noisy signals, on a tiny energy budget, whereas backpropagation needs globally orchestrated error transport and

runs on deterministic hardware that spends much of its energy suppressing noise. The fourth is agency: brains plan and act on an internal model of the world, which in AI is still largely engineered on top.

Brains show that all of this is physically achievable, and neuroscience has identified many of the mechanisms involved, from local plasticity [Bi and Poo, 1998] and replay [Wilson and McNaughton, 1994] to attractor dynamics [Khona and Fiete, 2022]. What AI lacks is therefore not an existence proof but a principled foundation: an account of why these mechanisms work that is general enough to carry over to other substrates.

1.2 Lost across scales

Few objects in science have been described at as many scales as the brain: molecules and synapses, neurons and circuits, regions and whole-brain networks, each with its own methods and its own models. What is still missing is an agreed and testable account, at the level of physics, of how living matter can infer and learn at all, and why similar capacities recur at very different scales, from single cells to whole brains [Lyon, 2005, Levin, 2019].

One symptom is that models at different scales rarely connect. Bottom-up approaches build whole-brain models from local biophysics, with neural masses, neural fields or mean-field approximations. They are mechanistic, but every modelling choice multiplies the number of plausible variants, and macro-scale data alone can rarely decide between them. Top-down approaches describe macro-scale phenomena, such as functional connectivity and resting-state networks, without a mechanism that generates them. The two research cultures rarely meet [Rosas et al., 2025]. A principle that holds at every scale would give both a common reference, against which bottom-up mechanisms and top-down descriptions could be tested.

1.3 Why existing frameworks do not close the gap on their own

Several frameworks supply parts of such a principle. None, we argue, supplies all of it.

- **Task-optimized deep networks (neuroconnectionism)** have given neuroscience its most successful models of sensory representations, and a productive way of comparing brains and machines on the same tasks [Richards et al., 2019, Doerig et al., 2023]. Their great strength is task fidelity: the models actually perform the functions they are meant to explain. Their neurobiological validity is partial, however. Learning is mostly left to backpropagation, units and dynamics are idealized, and the models are rarely generative models of whole-brain dynamics. More fundamentally, architecture, objective and learning rule are chosen by the modeller and judged by their fit to data. The approach can show which functions a brain-like system can learn, but it does not set out to explain why suitably organized matter comes to infer and learn in the first place.
- **Biophysical whole-brain models** sit at the other end of the spectrum. They are mechanistic and interpretable at their chosen scale, and they are the most direct link between large-scale dynamics and physiology [Deco et al., 2008, Sanz Leon et al., 2013, Breakspear, 2017]. But they have little task fidelity: they are not designed to learn or to perform the functions whose neural basis they describe, which also limits what they can offer AI. They are also generically bottom-up, taking the local mechanism as given at one privileged scale. The approach proposed here is meant to complement them with a scale-agnostic perspective (Section 4.2), not to replace them.
- **Physical learning, energy-based models and other local-rule systems** show how far memory, computation and learning can go with local interactions. Examples are Hopfield networks and Boltzmann machines [Hopfield, 1982, Ackley et al., 1985], dense associative memories with very large capacity and a close link to attention in transformers [Krotov and Hopfield, 2016, 2020, Ramsauer et al., 2020], equilibrium propagation and coupled learning in physical networks [Scellier and Bengio, 2017, Stern et al., 2021], and classical and neural cellular automata [Wolfram, 1984, Mordvintsev et al., 2020]. In each, however, the energy function, the update rule or the training objective is chosen by the designer, often for capacity or for supervised tasks, and few of these models are tied to brain data.
- **Large-scale attractor and energy-landscape models of brain dynamics** treat whole-brain activity as movement on a landscape, in a tradition that goes back to Amit [1989]. They include multistable and ghost-attractor regimes of connectome-based mean-field models [Deco and Jirsa, 2012, Golos et al., 2015], coupled-oscillator models [Cabral et al., 2011], pairwise maximum-entropy (Ising-type) landscapes fitted to fMRI [Watanabe et al., 2013, Ezaki et al., 2017, Gu et al., 2018], Hopfield-type networks built from mouse cortical connectivity [Fasoli et al., 2026] or estimated from

MEG [Gosti et al., 2024], attractor-state perspectives on the default mode network [Poerio and Karanagiotidis, 2025], and our own connectome-based attractor networks of the human brain [Englert et al., 2026]. These models capture real features of brain dynamics, but most are phenomenological: couplings are taken from connectivity or fitted to data, and dynamics and storage rules are chosen rather than derived. Why brain dynamics should take this form, and what the attractors compute, is largely left open.

- **The free energy principle and active inference** offer the broadest first-principles account of self-organization as inference [Friston, 2010, Friston et al., 2023a]. Their route is top-down. One starts from a whole system that persists, separated from its environment by a Markov blanket, and shows that its internal states can be read as encoding beliefs about external ones. The principle itself, like any principle, is not falsifiable; it is a heuristic to build on. Its technical foundations have been examined closely in recent years, and the critical literature has made clear which conditions the inferential reading requires and how hard they are to verify for real systems such as brains [Biehl et al., 2021, Aguilera et al., 2022, Bruineberg et al., 2021].

Each of these frameworks offers part of what is needed: models that perform tasks, mechanisms grounded in physiology, learning rules native to physics, or a first-principles language. What is missing is a construction that connects them: derived from stated physical assumptions, valid across scales, able to learn and perform tasks, and testable in brains and machines alike.

1.4 A different route: start from the simplest unit

The strategy of this programme is to invert the direction of the FEP argument. Instead of asking what any persisting system must look like, we ask what is minimally required for a piece of matter to infer and to learn, and we answer that question for the simplest building block first. There are four reasons for this choice.

The first is that physics has often made progress this way. The ideal gas, the single spin and the hydrogen atom are idealizations, but solving them exactly gave the vocabulary in which interacting systems could later be understood. Hopfield [1982] followed the same strategy for memory: simple units, and collective computation as an emergent property of their interactions.

The second is that assumptions about a unit can be checked. Whether a substrate has bounded coordinates, a separation of timescales or a well-defined reference state can be measured, and each of these can fail. A result proven for the unit under such assumptions holds for any substrate that meets them, biological or engineered. Conditions stated for a whole brain, such as the existence of a steady state and a Markov blanket, are much harder to verify.

The third is that composition becomes a hypothesis rather than a premise. Whether networks of units still infer and learn, and how, is then a separate question that can be answered by derivation and simulation (Section 4) rather than being assumed at the outset.

The fourth reason concerns scale. If a coarse-grained coordinate of a network satisfies the same assumptions as the unit, it is itself a unit of the same family. This is what licenses one model family at every level of description, and with it the fitting of these models to coarse measurements such as fMRI (Section 4.2).

If the construction holds, inference and learning become properties of a stated class of physical systems, rather than of algorithms or of systems that merely persist. The inference and learning parts of the FEP are then recovered for this class as consequences, without its general claims, which could give the principle a constructive and falsifiable core. The characteristic signatures of the resulting networks, such as priors stored as attractors, approximately orthogonal attractors and noise that encodes precision, become predictions for any substrate that meets the assumptions. And if the construction fails somewhere, the failure is informative, because it points to the assumption that broke.

1.5 Scope: the roots of adaptive behaviour, not its full complexity

We do not doubt that the brain’s complexity will remain a major challenge for a long time. Its many cell types, its molecular machinery, its developmental and evolutionary history and its individual variability will not be derived from a handful of assumptions. Our claim is narrower: the roots from which this complexity grows can be understood much better than they are now. Behind the many implementations that biology shows, from molecular networks to neural circuits, there may be shared principles of organization that each of them approximates in its own way.

One possibility, which we find plausible but state as a hypothesis, concerns how evolution relates to these principles. Evolution may not have had to invent inference and learning from scratch, as an assembly of ad hoc rules. It may instead have started from a property that suitably organized matter already has, if only weakly and imperfectly when assembled at random: slow structural coordinates that, once exposed to a structured environment, tend to align with it (Section 3). Evolution would then have found this property rather than devised it, and spent its effort on substrates that exploit it ever better [Fields and Levin, 2020, Levin, 2023]. From a different starting point, Fields et al. [2021] and Fields and Levin [2022] have argued that inference-like behaviour is generic to physical systems that interact with an environment.

The same scope shapes the predictions. Most of them are not claims about the brain’s full machinery but about the signatures of the model class that grows from the proposed root: priors stored as attractors, approximately orthogonal attractors (self-orthogonalization), noise that encodes precision, asymmetric couplings that carry activity along the landscape rather than down it (self-solenoidization), the timing of a predictive-coding learning signal, and an inferential form that survives coarse-graining (Sections 4 and 5). If evolution has settled on this class of solutions, these signatures should be found broadly, at different scales and in different substrates. For neuroscience they are markers to look for (Section 7), and their absence in systems that meet the assumptions would count against the programme. For AI, they point to the properties worth building into machines, rather than biological details worth copying.

1.6 How to read this document

Section 2 summarizes what is already established. Section 3 states the construction of the minimal unit, and Section 4 composes such units into networks. Section 5 asks what attractors are, and Section 6 collects biology-like phenomena that may follow. Section 7 lists the predictions with their failure criteria, Section 8 the technical plan, and Sections 9 and 10 the risks and the relation to other approaches. Section 11 collects further-reaching ideas on which nothing else depends. How the document came about, and how to comment on it or contribute, is described in *About this document: its origins, status and how to contribute*.

To keep the status of each statement visible, key claims carry a tag:

Tag	Meaning
PUBLISHED	Shown in peer-reviewed work, mainly Spisak and Friston [2026] and Englert et al. [2026]
IN PREPARATION	Derived or simulated in work being prepared for publication
HYPOTHESIS	A heuristic expectation, partly supported by simulations or data, that the programme will formalize and test
SPECULATIVE	An appealing but distant possibility, flagged so that it is not mistaken for a claim

2. Where we stand: what is already established

The programme does not start from zero. Two peer-reviewed papers form its published core, one theoretical and one empirical, and they show that the networks at the centre of this white paper exist, work and can be fitted to human brain data.

In theory and simulation **PUBLISHED**. Spisak and Friston [2026] derived a class of stochastic attractor networks from the FEP, by applying it recursively to a network of nested subsystems, without hand-designing either the inference or the learning rule. Each node infers by combining its own bias with weighted evidence from the other nodes, and each coupling learns by contrasting observed with predicted correlations, a local, predictive-coding rule. The paper showed, analytically and in simulations, that the attractors of such networks act as Bayesian priors, that the learning rule drives them towards approximate orthogonality even for correlated inputs, that the networks generalize to new inputs, that sequential input produces asymmetric couplings and directed sequence dynamics, and that the networks resist catastrophic forgetting. The simulation code is open.

In the human brain **PUBLISHED**. Englert et al. [2026] built networks of this class directly from human functional connectivity. Their attractors were approximately orthogonal, held up against strict null models, and

provided initial evidence that the landscape corresponds to canonical large-scale brain networks, replicates across datasets, and captures task-related changes and altered dynamics in clinical conditions. Independently, a connectome-based Hopfield-type model of mouse cortex predicts fMRI co-activation patterns from its attractor landscape [Fasoli et al., 2026]. Our open-source package *attractome* makes such fits available to others, and first fits in individual participants are under way (Section 8.2).

What is not yet established. The 2026 derivation starts from the FEP and from a description of the network as nested subsystems. It shows that if a system can be described in this way, these inference and learning rules follow. It does not show which physical conditions a piece of matter has to meet to behave as one of the nodes, and it inherits the open questions about the FEP’s assumptions (Section 1.3). Closing this gap from below is the step from which the rest of the programme proceeds. It is the subject of Section 3, and it is where most of the theoretical work of Line A lies. Equally open are the extension to hierarchy and action, the predictions that go beyond orthogonality, and the implementation on physical hardware.

3. A minimal inferring and learning unit

This section states the construction of the minimal unit and what follows from it. The aim is a sufficient route from physics to inference and learning, not a necessary one: a minimal and physically plausible model of how matter can be organized to infer and learn. The full derivations are in the manuscript “From physical dynamics to perception and learning” (Spisak et al., in preparation; available on request) **(IN PREPARATION)**. Here we give the assumptions, the results in informal form, and the known cases they reduce to. The notation, the ordering of timescales and a map of which result uses which assumption are collected in *Setting, notation and the assumption–result map*.

3.1 Why this is possible now

That inference might be a property of matter, and not only of minds, is an old intuition. Several developments, some long established and some recent, now fit together and make it possible to turn it into a construction.

A geometry in which prediction errors are natural objects. The non-parametric exponential manifold of Pistone and Sempi [1995] describes the space of all normalized probability densities of a system, without fixing a parametric model in advance. On this manifold every tangent vector is centred: it has the form “actual minus expected”. Prediction errors, which Bayesian brain theories postulate, can therefore be read as the generic way a normalized ensemble changes.

Learning as constrained maximum entropy. Jaynes’ maximum-entropy principle [Jaynes, 1957] and the duality theory of exponential families [Amari, 2016] relate each parameter of a distribution to the expectation it controls. Recent work has made the correspondence between constrained maximum entropy and the FEP explicit [Ramstead et al., 2023, Sakthivadivel, 2022]. This is what allows slow structural coordinates, such as couplings, to be treated as thermodynamic conjugates of correlations.

Timescale separation as a tool. Averaging and adiabatic elimination [200, 2008, Haken, 1983] turn a separation of timescales into a precise statement about which variables can be treated as fixed, which as equilibrated, and which as slowly drifting. In the construction this separates perception from learning.

A sharper question. The critical debate about the FEP [Biehl et al., 2021, Aguilera et al., 2022, Bruineberg et al., 2021] has clarified what a physical account of inference must specify. A construction whose assumptions are local and explicit is a well-posed contribution to that debate, rather than a rival grand theory.

A known target. Finally, the network rules of Spisak and Friston [2026] are now published. A construction from below has to recover them, including the exact form of the learning rule and its timing. Until recently, there was no comparably concrete target against which a unit-level construction could be checked.

3.2 Assumptions

The construction uses a small set of explicit assumptions, chosen by Occam’s razor: whenever several structures could support a result, we take the least elaborate one.

1. **A normalized reference ensemble (A1).** The stationary densities of the system are positive and regular enough to admit local exponential coordinates, and environmental forcing acts as an exponential tilt of a reference state.

2. **Coarse observables with a double separation of timescales (A2).** Microscopic variables relax fastest, retained mesoscopic coordinates next, and slow structural coordinates (couplings) much more slowly. The environment varies on an intermediate timescale.
3. **Bounded coordinates and a point of view (A3).** Retained coordinates are bounded, as in any finite physical system. One of them, the exposed coordinate, is followed from its own point of view, and the others form its boundary. No Markov blanket is required: shielding from the environment is a special case, not a precondition.
4. **Minimal interaction and least-structured response (A4).** Near a generic operating point, only the leading interaction between boundary and exposed coordinate is kept, and the resting law of the exposed coordinate is the maximum-entropy law on its range.
5. **Causal renewal, for networks (A5).** Coupled units renew together, each forming its prediction from the preceding state of its boundary.

Each assumption is an empirical question for a given substrate, and each can fail.

3.3 Results

The results are stated informally; precise statements, conditions and proofs are given in the manuscript in preparation. For each result we name the assumptions it uses and, in one line, the route by which it is obtained.

- **R1. Bayesian structure from normalization (A1, A2).** Normalizing the tilted ensemble over fast and slow coordinates turns the reference state into a prior, the tilt into a likelihood and the normalizer into model evidence. No beliefs are assigned to microstates. *Route:* the normalization of an exponentially tilted density, read at fixed slow coordinates.
- **R2. Prediction error as a tangent (A1–A3).** Along the fast direction, the centred tangent of the exposed coordinate is its prediction error, $\epsilon = \mu - \mathbb{E}[\mu \mid s, W]$. It follows from preserving normalization alone. *Route:* the tangent space of the non-parametric exponential manifold.
- **R3. The response function from boundedness (A1, A3, A4).** The unit’s mean response is the derivative of the log-partition function of the bounded maximum-entropy family: the Langevin function $L(x) = \coth x - 1/x$, which reduces to $\tanh$ for a binary coordinate. No activation function is chosen by hand. *Route:* Legendre duality of the bounded maximum-entropy family.
- **R4. Perception as fast relaxation (A1–A4).** The fast relaxation of the mesoscopic coordinates moves the distribution of the unit’s state towards the posterior; it is descent on variational free energy at fixed structure. *Route:* adiabatic elimination of the microscopic variables.
- **R5. Learning as slow relaxation (A1–A4; timing from A5).** Timescale separation makes the expected correlations of the mesoscopic coordinates the Legendre duals of the slow structural coordinates. The structure then climbs the average evidence, driven by a contrastive learning signal: observed minus expected correlation, a Hebbian minus an anti-Hebbian term. *Route:* a dynamical form of Jaynes’ constrained maximum entropy, by averaging over the fast timescale.
- **R6. Uncertainty from fluctuations (A1–A4).** At both timescales, fluctuations sample the corresponding posterior. For the fast state, variance equals gain, a fluctuation–dissipation relation; for the slow structure, this is the idea of synaptic sampling [Kappel et al., 2015]. *Route:* the second derivative of the log-partition function, which is both a variance and a susceptibility.

The two centred quantities that drive perception and learning can be written compactly:

The construction in two sentences

Perception. A prediction error, geometrically, is nothing more than a normalization-preserving, and therefore expectation-centred, tangent vector on a non-parametric statistical manifold.

$$\varepsilon_\mu = \underbrace{\mu}_{\text{actual state}} - \underbrace{\mathbb{E}[\mu \mid s, W]}_{\text{expected state (prediction)}} \quad (1)$$

Learning. A synapse is the thermodynamic Legendre conjugate of a correlation.

$$\psi(W) = \sup_q \left\{ \underbrace{W}_{\text{structural parameter}} \underbrace{\mathbb{E}_q[s\mu]}_{\text{expected correlation}} - \underbrace{D_{\text{KL}}(q \parallel q_0)}_{\text{relative entropy}} \right\} \quad (2)$$

$$\varepsilon_W = \underbrace{s\mu}_{\text{observed correlation}} - \underbrace{\mathbb{E}[s\mu \mid W]}_{\text{expected correlation}} \quad (3)$$

Here ψ is the log-partition function (a free energy) and q_0 the reference law of the unconstrained dynamics. The structural parameter and the expected correlation form a Legendre pair, with relative entropy as the dual potential. Since $\partial\psi/\partial W = \mathbb{E}[s\mu \mid W]$, the learning error is the gradient of the dual objective $W \cdot s\mu - \psi(W)$: learning climbs it until expected and observed correlations agree, which is Jaynes' constrained maximum entropy in dynamical form. Synapse is meant in the general sense: any slow coupling between units.

3.4 Known cases, and what the unit does not do

A new construction should reduce to known results where those apply. For binary units, symmetric couplings and equilibrium, R3 gives the familiar tanh response of Ising-type models, and R5 gives a contrastive, Hebbian minus anti-Hebbian rule of the same form as Boltzmann-machine learning [Ackley et al., 1985]. For a continuous bounded coordinate, R3 gives the Langevin function of the classical paramagnet. R6 is the fluctuation–dissipation relation of the exponential family [Kubo, 1966]. What is new is not these ingredients but their joint derivation from one set of physical assumptions, their Bayesian reading, and the timing of the learning signal. In the construction this timing follows from causal renewal (A5), and in the networks of Spisak and Friston [2026] it is predicted to be what makes the attractors orthogonalize (P3).

Just as important is what the construction does not claim. It shows one route from physics to inference and learning, not the only one, and it does not show that brains, or any particular material, take this route. Where its assumptions fail, for example without a clear separation of timescales, the inferential reading is not guaranteed, and the learning signal becomes biased. The unit is one-dimensional, so multidimensional or spiking units need extensions.

The construction also defines a *family* of units and networks, not one set of equations. Three ingredients come from the substrate and select a member of the family: the reference measure of the microstates, the bounds of the retained coordinates (which fix the response function, the Langevin function in general and tanh only for a binary coordinate), and the type of intrinsic interaction between mesoscopic coordinates. What all members share is the Bayesian structure, the centred prediction error, the Legendre-dual learning signal and the signatures that follow from them. When this white paper speaks of the same rules at different scales or in different substrates, it means rules of this family.

Nor does the construction require finely tuned structure. Structural coordinates may start far from any useful configuration; the slow dynamics then move them until expected and observed correlations agree, so that the structure aligns itself with the statistics it is exposed to. How far this self-alignment can go is limited by the substrate, by the range, precision and timescales of the structural coordinates it offers. A physical system may thus obey the construction only weakly, with little headroom, and evolution and development can be read as processes that build substrates with more headroom (Sections 1.5 and 11) (HYPOTHESIS).

Why “entropic”. The name refers to the dynamics, not only to their endpoint. A long tradition treats relative entropy as what drives motion, not just as a description of where a system ends up: fluxes driven by entropy gradients in irreversible thermodynamics [Onsager, 1931], the Fokker–Planck equation as a gradient flow of relative entropy [Jordan et al., 1998], and the split of dynamics into a reversible, energy-driven part and an irreversible, entropy-driven part [Grmela and Öttinger, 1997], which mirrors the decomposition used in Section 4.5. In this sense, inference in the units is a relaxation that lowers relative entropy towards the posterior, learning is constrained maximum entropy, and the solenoidal part of the dynamics leaves relative entropy unchanged. As with entropic forces, what drives all three is the multiplicity of hidden microstates, not a designed interaction. The programme of Entropic Dynamics [Caticha, 2011] shares this intuition but not the construction.

Setting, notation and the assumption–result map

Setting. A physical system has microscopic variables that are not tracked individually. A small number of coarse observables over them are retained: bounded mesoscopic coordinates, one of which, the exposed coordinate μ , is followed from its own point of view, while the others form its boundary s . Slow structural coordinates W (couplings) parametrize the interaction between boundary and exposed coordinate. The environment acts on the system as an exponential tilt of a reference ensemble q_0 .

Timescales. The construction uses the ordering

$$\tau_{\text{micro}} \ll \tau_{\mu} \ll \tau_{\text{env}} \ll \tau_W \quad (4)$$

The first inequality allows the microscopic variables to be eliminated (perception, R4). The middle one means that the environment changes slowly enough for the state to track it, but fast enough for the structure to average over it. The last allows the structure to be averaged over the fast and intermediate timescales (learning, R5).

Symbol	Meaning
μ	Exposed mesoscopic coordinate (a node’s state), bounded
s	Boundary coordinates (the states that the exposed coordinate interacts with)
W	Slow structural coordinates (couplings); J in networks
b	Bias: the unit’s own field, read as a prior
q_0	Reference law of the unconstrained dynamics
ψ	Log-partition function (a free energy)
L	Langevin function, the mean response of the bounded maximum-entropy family; tanh for a binary coordinate
$\varepsilon_{\mu}, \varepsilon_W$	Prediction error of the state and learning signal of the structure
D_{KL}	Relative entropy (Kullback–Leibler divergence)

Assumption–result map. A filled cell means that the result uses the assumption.

	A1 Reference ensemble	A2 Timescales	A3 Bounds, point of view	A4 Minimal interaction	A5 Causal renewal
R1 Bayesian structure	•	•			
R2 Prediction error	•	•	•		
R3 Response function	•		•	•	
R4 Perception	•	•	•	•	
R5 Learning	•	•	•	•	timing
R6 Fluctuations	•	•	•	•	
Networks (Section 4)	•	•	•	•	•

4. From units to networks: entropic attractor networks

Composed in a straightforward way, the minimal units form entropic attractor networks (eANNs): stochastic attractor networks whose attractors are Bayesian priors and whose learning rule self-orthogonalizes them [Spisak and Friston, 2026]. These are the networks of Section 2, now obtained from below rather than from the FEP.

4.1 Inference and learning rules

When units are coupled, each unit’s boundary consists of the other units’ states. Each node then performs local Bayesian inference, combining its bias (prior) with weighted evidence from the other nodes through the Langevin function L , and adapts its couplings with a local rule that contrasts observed with predicted correlations:

$$\mathbb{E}_q[\sigma_i] = L\left(b_i + \sum_{j \neq i} J_{ij} \sigma_j\right), \quad \Delta J_{ij} \propto \sigma_i \sigma_j - L\left(b_i + \sum_{k \neq i} J_{ik} \sigma_k\right) \sigma_j \quad (5)$$

The first term of the learning rule is Hebbian (observed correlation), the second anti-Hebbian (predicted correlation): a predictive-coding rule. It belongs to the family of contrastive, local learning rules whose relation to inference has been worked out for predictive-coding networks [Millidge et al., 2022] and, in the symmetric equilibrium case, for Boltzmann machines [Ackley et al., 1985]. The network is a generalized, stochastic Hopfield network, a NESS-generalization of Boltzmann machines, whose noisy dynamics amount to Markov chain Monte Carlo sampling from the global posterior. The construction in Section 3 fixes the timing of the rule: the prediction is formed from the preceding network state and paired with the renewed one. In network terms, the two sentences of Section 3.3 become concrete: each node’s prediction error is the centred tangent of its state, and each coupling J_{ij} is the thermodynamic Legendre conjugate of the correlation between nodes i and j , which it adjusts until observed and predicted correlations agree.

One-dimensional units, binary questions. The minimality assumption makes every node, and therefore every attractor, one-dimensional: its continuous-Bernoulli state expresses graded evidence for or against a single event. This is why, with zero biases, the sign-inverse of every attractor is also a fixed point. An attractor and its mirror image are one prior, read as evidence for or against the same latent cause.

4.2 Across scales: one model family and the same landscape at every resolution

Both the generality of the approach and its application to fMRI rest on what survives coarse-graining: the form of the construction, and the attractor landscape it produces. Partly IN PREPARATION HYPOTHESIS for brain data.

One family of forms. Timescale separation is used twice: within the unit, where it separates perception from learning (Section 3.3), and between levels, where fast modes are enslaved by slow ones that carry the long-run dynamics [Haken, 1983, Carr, 1981, Kuramoto and Nakao, 2019, Jafarian et al., 2021]. In whole-brain models, this has been proposed, and demonstrated in simulations, as the mechanism by which low-dimensional structure emerges [Runfola et al., 2025]. If a coarse unit, such as a column or a brain region, exposes one macro-observable, a conjugate field acting on its transition ratios again yields an exponential family, with a response function and a susceptibility equal to its variance. The particular reference family need not survive averaging, but the architecture does: one exposed statistic, a conjugate field, fluctuation–dissipation, and a Bayesian reading when the macrostate represents an external cause. What changes with substrate and scale is which member of the family applies: the reference measure of the microstates, the bounds of the retained coordinates (and with them the shape of the response function, tanh only in the binary case), and the type of intrinsic interaction between mesoscopic coordinates. Single neurons, cortical columns and brain regions are therefore not expected to obey identical equations, but rules of one family, which share its signatures. Rules of this family can therefore apply from single neurons, where the learning rule resembles spike-timing-dependent plasticity [Bi and Poo, 1998] HYPOTHESIS, to brain regions (criterion D2, Section 10). Making this rigorous for networks is a goal of Line A (A2).

The same landscape. An fMRI parcellation does not create a new network. It reports the activity of one neural system in fewer coordinates, the parcel means $y = C\sigma$: it is a camera, and different atlases are different cameras pointed at the same scene. The attractors $p\mu$ of the fine system are whole-brain patterns. Writing each as its regional profile $a\mu = Cp\mu$ plus a within-parcel residual $r\mu$, the one essential assumption

is homogeneity: on the attractors that matter, units within a parcel participate alike, which is exactly what functional parcellations are designed for [Gordon et al., 2014, Eickhoff et al., 2018]:

$$p_i^\mu = a_{\alpha(i)}^\mu + r_i^\mu, \quad \|\mathbf{r}^\mu\| \ll \|\mathbf{p}^\mu\| \quad (6)$$

Because the residuals average to zero within each parcel, fine inner products split exactly into a size-weighted coarse inner product and a residual term that homogeneity makes small:

$$\langle \mathbf{p}^\mu, \mathbf{p}^\nu \rangle = \sum_{\alpha} |P_{\alpha}| a_{\alpha}^{\mu} a_{\alpha}^{\nu} + \langle \mathbf{r}^\mu, \mathbf{r}^\nu \rangle \quad (7)$$

Three consequences follow. First, attractors that differ across parcels remain visible, those that differ only within parcels merge, and nothing new should appear. In nested atlases [Urchs et al., 2019], higher resolution splits coarse attractors into descendants that average back to them. Second, approximate orthogonality at the fine scale is inherited at the coarse scale, in the size-weighted parcel metric (analyses in unweighted or z-scored coordinates therefore use size-balanced atlases). Third, only a functional camera is useful: spatially scattered random parcels average structured patterns away, shrinking them roughly with the square root of parcel size until at most a global mode remains, whereas contiguous random parcels degrade gradually, the finest attractors first. The claim is not that a network can be fitted at any scale, but that meaningful coarse-grainings preserve the landscape and meaningless ones erase it [Meshulam et al., 2019, Friston et al., 2021, Machta et al., 2013, Transtrum et al., 2015].

Assumptions, limits and tests. Besides homogeneity, detecting the landscape with fitted networks assumes that the slow dynamics are organized by a modest number of attractors, and that the coarse network models the observed regional process. Orthogonality is a prediction (P1, P2), not a prerequisite. Nothing assumes that a parcel is a large neuron, that couplings survive block averaging, or that coarse-graining is an exact renormalization map. After haemodynamic deconvolution, fMRI is a slow camera for slow patterns [Friston et al., 2014, Heitmann and Breakspear, 2018]. Averaging cannot recover distinctions it has discarded, and residual haemodynamic mixing can mimic mixtures of successive states, which simulations must rule out before such mixtures are read as prior mixing (P10). Nested atlases in population and precision fMRI [Gordon et al., 2017] make the argument directly testable: project fine attractors onto coarse atlases and match them; check that coarse attractors split into descendants; measure homogeneity as the between-parcel share of each attractor’s norm; compare inherited inner products with dimension-matched nulls; and run random-parcellation controls. The resolution range is fixed in advance by test–retest reliability, and matching is repeated within single participants (B1; P18–P20).

In one sentence

Fit models of the same family at every resolution, and the attractors should be the same objects, seen through a coarser or finer camera.

4.3 Emergent properties

- **Attractors as priors** (PUBLISHED). Attractor states store the network’s priors, and ongoing activity combines them with current input into a posterior. Local inference at the nodes gives rise to inference at the level of the whole network.
- **Self-orthogonalization, the key property** (PUBLISHED) theory, in-silico demonstrations, first evidence in human fMRI. Even for correlated inputs, the predictive-coding rule drives attractors towards approximate orthogonality: a new pattern is stored as its residual from what is already known. The network thereby approximates a Kanter–Sompolinsky projector network [Personnaz et al., 1986, Kanter and Sompolinsky, 1987], whose attractors coincide with the leading eigenvectors of the coupling matrix, but with a local learning rule. Offline routes towards similar couplings are known as well: Hebbian unlearning [Hopfield et al., 1983] and “dreaming”, which provably drives Hebbian couplings towards the projector [Fachechi et al., 2019] and can exploit correlations in the data [Serricchio et al., 2025]. In effect, self-orthogonalization is an on-line, approximate principal component analysis, in the tradition of Hebbian and anti-Hebbian subspace learning [Oja, 1982, Földiák, 1990, Pehlevan and Chklovskii, 2019]. Orthogonal attractors can act towards separating latent causes and use storage capacity efficiently. Furthermore, they serve as a basis for the emergent capabilities discussed below.

- **Attractor–prior duality** Symmetric case: formally established equivalence; antisymmetric extension **(HYPOTHESIS)**. This is the Hopfield network–restricted Boltzmann machine equivalence [Barra et al., 2012] in disguise. Approximate orthogonality makes the network statistically equivalent to a two-layer network in which the attractors are actual hidden nodes, connected only to the original nodes. Our heuristic is that, with an antisymmetric coupling component, non-equilibrium steady states and solenoidal dynamics appear as within-layer connections between the attractor nodes. The appeal is direct: attractors act as integrator nodes that pool evidence from many regions and feed the result back, performing (Bayesian) information integration. A change in a single connection can then be decomposed into changes in its contribution to individual attractors.
- **Superposition and prior mixing** **(HYPOTHESIS)** observed in simulations. Linear combinations of stored priors can exist as attractors or as “ghost attractors”, regions of slow dynamics without a true minimum [Deco and Jirsa, 2012]. Such slow regions can do real work: in classical and dense associative memories, patterns remain transiently retrievable above capacity from the slow remnants of their former basins [Clark, 2026]. Unlike the spurious states of classical Hopfield networks [Amit et al., 1985], they are useful here: the network can span the whole subspace of learned patterns and generalize to new inputs. Dense associative memories can likewise combine exponential capacity with emergent, useful minima that are not stored patterns [Hoover et al., 2025]. Mixing can also unfold in time, as quasi-periodic alternation across several attractors whose time-averaged posterior matches the data. Self-orthogonalization is what makes this work, and it offers a model for oscillatory brain dynamics [Aitchison and Lengyel, 2016, Echeveste et al., 2020].
- **Self-solenoidization** **(HYPOTHESIS)** observed in simulations. With our emergent learning rule, sequential presentation yields asymmetric couplings [Sompolinsky and Kanter, 1986, Kleinfeld, 1986]. During transitions, the landscape set by the symmetric couplings tends to absorb the flow components along the free-energy gradient, via a mechanism closely related to self-orthogonalization. What remains is encoded mainly in the antisymmetric couplings, as a transversal flow along the free-energy isocontours: approximately solenoidal flow that replays sequences without changing the steady state (transversality derivation is in progress, full solenoidality is only a conjecture). Self-solenoidization would thus be an on-line, approximate Helmholtz decomposition, as self-orthogonalization is an on-line, approximate principal component analysis. It is the learned counterpart of potential–flux decompositions of non-equilibrium dynamics [Ao, 2004, Kwon et al., 2005, Wang et al., 2008, Yan et al., 2013]. Changing only the antisymmetric component may allow new action policies without altering the perceptual model (Section 4.5).
- **Continual learning** **(HYPOTHESIS)** initial demonstration **(PUBLISHED)**. Two processes support it. First, stochastic multistability causes spontaneous replay of stored patterns, especially but not only in sequence dynamics, which consolidates old memories, as hippocampal and human replay is thought to do [Wilson and McNaughton, 1994, Liu et al., 2019] and as generative replay does in machine learning [Shin et al., 2017, van de Ven et al., 2020]. Second, self-orthogonalization stores new information only as its residual relative to what is known, so old knowledge is preserved as long as it remains at least mildly useful. This parallels orthogonal-subspace methods in machine learning [Zeng et al., 2019] and orthogonal task representations in brains and networks [Libby and Buschman, 2021, Flesch et al., 2022], as well as attractor scaffolds in hippocampal models whose memory degrades gracefully rather than collapsing [Chandra et al., 2025]. Together they give resistance to catastrophic forgetting [McCloskey and Cohen, 1989, French, 1999] without task labels or explicit regularization [Kirkpatrick et al., 2017]. The same two processes should matter on long-horizon tasks, where a system has to keep learning while it acts, without eroding what it learned earlier in the task (Section 1.1; P23).

4.4 Noise as information

In these networks, noise is not a nuisance to be suppressed. Its size encodes the precision of what a unit represents **(IN PREPARATION)** and partly already **(PUBLISHED)**.

Each unit’s fluctuations sample its posterior, so their variance is the posterior variance. By the fluctuation–dissipation relation of the exponential family [Kubo, 1966], the same quantity is the unit’s susceptibility, or gain. A unit is therefore most responsive where it is most uncertain, and least responsive where it is saturated and certain. Uncertainty is physically available to the system twice: as the size of its fluctuations and as its gain. This is the fluctuation–dissipation theorem put to work for computation.

In information-geometric terms, this quantity is the Fisher metric [Amari, 1998, 2016]. It is the metric in which prediction-error tangents are measured, so precision weighting is simply the geometry of that tangent space. At the network level the relevant metric is shaped by learning, through the couplings, and lowering indices with it cleanly separates the gradient and rotational parts of the dynamics (Section 4.5). In the minimal picture, the metric supplies the precision weighting that predictive coding requires [Feldman and Friston, 2010], without a separate mechanism for estimating precision: the inverse of a unit’s predictive variance is the precision of its prediction. Finite samples from a distribution produce exactly this kind of noise, so a system that learns from finite experience represents its uncertainty without extra machinery.

This view connects to sampling-based accounts in which neural variability represents uncertainty [Hoyer and Hyvärinen, 2003, Fiser et al., 2010, Buesing et al., 2011, Orbán et al., 2016], to evidence that gain variability tracks uncertainty in visual cortex [Hénaff et al., 2020], and to work on the sources and benefits of noise in neural systems [Faisal et al., 2008, McDonnell and Ward, 2011]. It is also what makes the framework a natural fit for stochastic and thermodynamic computing, which treat physical noise as a resource rather than an error [Camsari et al., 2019, Coles et al., 2023, Melanson et al., 2025].

4.5 Multistability, metastability and intrinsic action

Two dynamical properties of the networks matter both for brain dynamics and for action. Multistability comes from the attractors, and metastability from what the network learns about the transitions between them.

Multistability (PUBLISHED). Multiple attractors make the networks multistable, as theory, simulations and our brain models consistently show. Noise is what makes this multistability dynamic: the system visits several attractors over time instead of settling in one, as has been proposed for resting-state activity [Deco et al., 2010, Deco and Jirsa, 2012]. A deformable attractor manifold has recently been fitted to resting-state fMRI in the same spirit [Athanasiadis et al., 2025].

Metastability (HYPOTHESIS) observed in simulations. Sequence learning produces asymmetric couplings and a non-equilibrium steady state, with transient dwelling near attractors and directed transitions between them [Kelso, 1995, Rabinovich et al., 2008, Tognoli and Kelso, 2014, Hancock et al., 2024]. Because of self-orthogonalization, we expect these dynamics to be dominated by approximately solenoidal flow over the scalar free-energy landscape (self-solenoidization, Section 4.3). Broken detailed balance in human brain dynamics [Lynn et al., 2021, Kringelbach et al., 2024] is consistent with this picture, and the non-equilibrium thermodynamics of networks with asymmetric couplings can be characterized analytically in kinetic Ising models [Aguilera et al., 2023]. In cortex, expectation modulates metastable attractor dynamics and speeds up coding [Mazzucato et al., 2019], close to the expectation-as-prior account tested in P6.

Intrinsic action: paths over a learned world model (HYPOTHESIS). The intuition is that the network learns paths over a world model it has already learned. Variational free energy (VFE) describes perception on the scalar landscape. A related quantity, possibly the expected free energy (EFE) of active inference [Friston et al., 2015, Parr et al., 2022], would describe intrinsic action, carried by the antisymmetric couplings between pairs of nodes. These couplings induce quasi-solenoidal dynamics: paths along isocontours of the landscape that determine the transitions. Without input, the same dynamics would produce spontaneous fluctuations at rest, or something like dreaming [Hobson and Friston, 2012], as in the spontaneous replay of Spisak and Friston [2026]. Whether the standard EFE arguments carry over to these networks is an open question [Millidge et al., 2021].

Decisions and planning (HYPOTHESIS). Learned transition paths could support decisions based on learned data, and planning as inference [Attias, 2003, Botvinick and Toussaint, 2012], especially across multiple timescales [Kiebel et al., 2008], in the spirit of predictive maps and replay-based planning [Pfeiffer and Foster, 2013, Stachenfeld et al., 2017]. This has a continual-learning side too: the perceptual model, or world model, need not be relearned for every new task. Only the transition paths across its attractors change.

A local sketch. The mean-field dynamics and their linearization around a fixed point make the split between gradient and rotation explicit:

$$\dot{\theta}_i = -\gamma \left[\theta_i - b_i - \sum_j J_{ij} L_j(\theta_j) \right], \quad \delta \dot{\theta} = M \delta \theta, \quad M = -\gamma (I - JD), \quad D = \text{diag}(\psi_1''(\theta_1), \dots, \psi_n''(\theta_n)) \quad (8)$$

Here D holds each unit's susceptibility, which is also its variance and its Fisher information (Section 4.4). Splitting the couplings into symmetric and antisymmetric parts, $J = S + A$, and lowering an index with the Fisher metric separates a symmetric, gradient-like term from an antisymmetric, rotational one:

$$DM = -\gamma(D - DSD) + \gamma DAD \quad (9)$$

For two units with $J_{12} = S + A$ and $J_{21} = S - A$, the eigenvalues are $-\gamma \pm \gamma\sqrt{(S^2 - A^2)} \psi_1'' \psi_2''$. When $|A| > |S|$, the local dynamics contain a damped rotation with frequency $\omega = \gamma\sqrt{(A^2 - S^2)} \psi_1'' \psi_2''$, which links asymmetry directly to oscillation. Symmetric couplings can admit a reversible, equilibrium model, whereas antisymmetric couplings generically obstruct that interpretation and support directed probability currents. The cautious conclusion: asymmetry is the minimal coupling structure that distinguishes temporal consequence from static correlation, and turns an attractor landscape into directed sequence dynamics.

Directed connectivity through the attractors [HYPOTHESIS](#). In the dual picture (Section 4.3), attractors act as hidden nodes, and learned sequences live in directed, antisymmetric couplings between them. If region-to-region asymmetries serve mainly the transitions between attractors, the antisymmetric part of the couplings is low-rank and confined to the attractor subspace. With the m attractor patterns as the columns of Ξ , the $N(N - 1)/2$ free asymmetries between region pairs reduce to a small matrix Ω of attractor-to-attractor flows, with about $m(m - 1)/2$ free entries. With Λ holding the precision (curvature) of each basin, a linear-Gaussian sketch with isotropic noise gives the condition under which this flow is solenoidal, that is, leaves the stationary density of the landscape unchanged [\[Kwon et al., 2005\]](#):

$$H = \Xi \Lambda \Xi^\top, \quad T = \Xi \Omega \Xi^\top, \quad \text{solenoidal} \iff \Omega = Q \Lambda, \quad Q^\top = -Q \quad (10)$$

Exact solenoidality therefore does not require attractors of equal depth, which the present models do not have. It requires each directed coupling to be weighted by the precision of the attractor it targets, $\Omega_{\mu\nu} = Q_{\mu\nu} \lambda_\nu$: an evidence weighting of the transitions. Besides their antisymmetric part, the transition couplings then carry a small symmetric part, $\frac{1}{2} Q_{\mu\nu} (\lambda_\nu - \lambda_\mu)$, which grows with the difference in depth. Whether the predictive-coding learning rule produces this weighting by itself, since it subtracts precision-weighted predictions, is open, and simulations can measure it directly. A second route appears at the level of metastable hopping between attractors. There, added probability currents leave the occupancies unchanged exactly when they decompose into closed cycles, each added transition rate being scaled by the inverse occupancy of its source [\[Schnakenberg, 1976\]](#). A directed flow between just two attractors cannot preserve occupancies, but a cycle through three or more can, whatever their depths. Both routes lead to the same testable signature: attractor occupancies set by the symmetric landscape alone, while the asymmetric couplings produce currents that leave those occupancies unchanged. Finding this despite unequal depths would itself be evidence of approximate self-solenoidization (P9).

An opportunity for directed connectivity from fMRI [HYPOTHESIS](#). Estimating directed connectivity between single pairs of regions from fMRI is notoriously hard, whether with dynamic causal modelling, Granger causality or related methods [\[Friston et al., 2003, Smith et al., 2011, Seth et al., 2015, Gilson et al., 2016\]](#), and the antisymmetric component is typically weak compared with the symmetric one. If asymmetries serve attractor transitions, estimating them at the attractor level pools evidence from many region pairs into a few attractor-to-attractor flows, which should improve statistical power. Where needed, the flows can be projected back onto region pairs as $\Xi \Omega \Xi^\top$ (P22). This applies the logic of the sufficiency test (Section 5.2) to directed rather than undirected structure, and it fits the coarse-grained state spaces in which broken detailed balance has been detected in human brain dynamics [\[Lynn et al., 2021\]](#). Haemodynamic delays that differ between regions can create spurious asymmetries, and they do not average out at the attractor level. Deconvolution, simulations with region-specific haemodynamic responses, and sequence tasks with a known order serve as controls.

5. Attractors as levels of description

From the observer to the system. In Section 4.2, coarse-graining is done by an observer. Attractors are, in effect, how a system coarse-grains itself: they create slow, macroscopic variables from within, and so create the levels that a camera can later observe.

5.1 Attractors are not epiphenomena

Attractors are often dismissed as epiphenomena. In this programme they are how new timescales, and with them new levels of description, come into being ([HYPOTHESIS](#)) builds on ([PUBLISHED](#)) results.

A common objection to attractor accounts of brain function is that attractors are convenient summaries of the dynamics, while the causal work is done by neurons and synapses alone. Pressed to its end, the objection becomes a philosophical question about causation across levels of description [[Kim, 1998](#)], which we do not claim to settle. The programme does, however, offer several concrete reasons for the opposite view, each of which can be tested:

- **Attractors create timescales.** An attractor is where the dynamics slow down: the system typically dwells near it far longer than any of its units takes to relax. It is therefore a new, slower variable that the system creates for itself, and this is exactly the kind of timescale separation from which Section 3 derives perception and learning. The state of an attractor network can then act as a slowly varying field, or prior, for faster processes, so that a construction of the same kind can repeat one level up (A2, Section 8.1). Attractors are thus not a by-product of inference and learning. They are how a system produces the separation of timescales on which its inference and learning rest. Hierarchies of intrinsic timescales across the cortex are consistent with this view [[Kiebel et al., 2008](#), [Murray et al., 2014](#)], as is the self-organization of functional hierarchies in recurrent networks with multiple timescales [[Yamashita and Tani, 2008](#)].
- **Attractors do computational work.** They can store priors and separate latent causes through self-orthogonalization; they may bridge delays in trace conditioning and working memory; and, through their transitions, they can carry sequences and intrinsic action (Sections 4.3–4.5 and 6). With symmetric couplings and approximately orthogonal attractors, the network is statistically equivalent to one in which the attractors are explicit hidden nodes that pool evidence and feed it back [[Barra et al., 2012](#)] (see Section 4.3). A description in which attractors are variables is then not a metaphor but an equivalent model.
- **Attractors make a difference under intervention.** On the interventionist account of causation [[Woodward, 2004](#)], a variable is a cause if intervening on it changes what follows. Perturbation experiments reveal attractor dynamics in exactly this sense. An activity bump imposed optogenetically on the fly’s heading circuit is maintained by the circuit [[Kim et al., 2017](#)]. Persistent activity in mouse frontal cortex returns to discrete attractor states after transient perturbations, or occasionally switches to the other attractor, followed by a switch in the animal’s action [[Inagaki et al., 2019](#)], and attractor dynamics gate the influence of distracting input on decisions [[Finkelstein et al., 2021](#)]. The programme’s own predictions are stated at the attractor level (extinction adds an attractor, chronic pain deepens one, expectations shift the posterior between them; P5, P6, P8) and can be probed with current neuroscience techniques, e.g. in closed loop with real-time fMRI (B6).
- **Attractors can form closed levels.** A macroscopic level is self-contained if it predicts its own future as well as the full microscopic state does (informational closure), and if interventions on macroscopic variables suffice to determine macroscopic outcomes (causal closure). [Rosas et al. \[2024\]](#) formalize such levels, which they liken to software running on physical hardware, and show that in a Hopfield network the overlaps with the stored patterns form a closed level of this kind: memory retrieval is computed at the level of the attractors, independently of microscopic detail [[Hoel et al., 2013](#), [Rosas et al., 2020](#)]. Timescale separation is what makes slow variables approximately closed: in synergetics, slow order parameters are generated by the fast variables and in turn enslave them, a circular causality in which the macro level is anything but idle [[Haken, 1983](#), [Kelso, 1995](#)]. Section 4.2 adds that the inferential form survives coarse-graining, so a closed level can itself infer and learn. Whether attractor-level descriptions of brain dynamics are closed in this sense is an empirical question (P17). If it is, closure has a practical corollary: for whatever is computed at that level, the attractor state is a sufficient statistic, which turns closure into a test of what is computed in the attractors (Section 5.2; P21).

Levels that can be studied in their own terms. [Simon \[1962\]](#) argued that complex systems evolve far more readily when they can build on stable intermediate forms, and that the result is a nearly decomposable hierarchy, in which fast interactions within subsystems are separated from slow, aggregate interactions between them. Attractors are dynamical counterparts of such stable intermediate forms. Where levels are approximately closed in this sense, each can be studied largely in its own terms, without simulating everything beneath it, and an elementary step that recurs at each level needs to be understood only once (Section 11).

5.2 What is computed in the attractors? Counterfactuals and a sufficiency test

For brain data, the programme aims at mechanism, not at predictive accuracy. Two kinds of test follow: counterfactual simulation, and a test, derived from closure, of which quantities are carried by large-scale attractors and which are computed below them ([HYPOTHESIS](#)) proposed method.

Mechanism, not accuracy. Data-driven predictive models of brain and behaviour are optimized to predict, and a mechanistic model need not beat them at it. Its value lies in answering questions they cannot answer: what would happen under an intervention that is absent from the training data. The fitted networks are generative and can be manipulated in silico: an attractor can be deepened or flattened, prior precision raised or lowered, the biases of selected regions shifted, or couplings altered. The decisive test is whether such counterfactual simulations, specified in advance, predict the effects of real manipulations, such as placebo and nocebo instructions, conditioning and extinction, pharmacological challenge, non-invasive brain stimulation or closed-loop real-time fMRI. The comparison is with mechanistically different variants of the same model (couplings fixed, biases only, a landscape without attractor structure), not with black-box predictors. Conventional predictive models keep an important role, however: as shown next, they set the reference that an attractor-level description has to reach, not beat.

Two levels in one fit. Score matching separates two components of the regional dynamics (graphical abstract, Line B). The couplings define the landscape and its attractors: dynamics at the network level. The biases capture what drives each region from below and from outside: mesoscale computation within regions, and sensory input. Both can be read out as time series from the same data: the attractor state $a(t)$, given by the overlaps of the regional state $x(t)$ with the fitted attractors (or by the attractor towards which each state relaxes), and the input $b(t)$ inferred by the fitted model.

Closure implies sufficiency. If the attractor level is closed in the sense of Section 5.1 (P17), and a quantity y (a perceptual report, a behavioural choice, an experimental condition or a clinical variable) is computed at that level, then the regional data carry no information about y beyond what the attractor state carries. The attractor state is then a sufficient statistic for y :

$$p(y | x) = p(y | a(x)) \iff I(y; x) = I(y; a) \quad (11)$$

Operationally, y should be predictable from the attractor time series just as well as from the full regional data: not better, but without loss. Because $a(t)$ and $b(t)$ are both computed from the regional time series by the fitted model, the chain rule of mutual information splits what the data carry about y exactly into three parts:

$$I(y; x) = \underbrace{I(y; a)}_{\text{attractor level}} + \underbrace{I(y; b | a)}_{\text{upstream input}} + \underbrace{I(y; x | a, b)}_{\text{neither}} \quad (12)$$

Quantities for which the first term exhausts the total are computed in attractor space. Those that need the second term are carried by upstream, mesoscale processing or by sensory drive. A substantial third term points to structure that the model misses. In this way closure, a property of a level, becomes a criterion for what lives on that level.

Mixed cases and controls. Many brain functions will be mixed, and the attractor share $I(y; a)/I(y; x)$ is then a graded measure rather than a verdict. Its value is that it is principled rather than descriptive: the criterion follows from the theory, not from choosing whichever feature set predicts best. Three controls keep it honest. First, sufficiency must be specific: equally compact coarse-grainings that ignore the attractors (principal components, random projections, canonical networks of the same dimension) should lose information that the attractor coordinates keep; otherwise the result reflects compression in general rather than attractors. Second, the information terms are estimated from the cross-validated performance of nested models fitted with the same learner, and are bounded in practice by sample size, model capacity and haemodynamic blurring. Equal performance is therefore established with equivalence tests, not read off a non-significant difference. Third, sufficiency for an external quantity is necessary but not sufficient for closure, which also requires the attractor level to predict its own future (P17), so the two tests are run together. The same logic applies between resolutions: replacing x with data from a finer atlas, or at voxel level, asks whether the coarse attractors suffice for what the finer data carry (Section 4.2).

An example: pain. Pain is a natural test case because it combines strong input with strong expectations. If expectations act as attractor priors (P6), placebo and nocebo effects on reported pain should be largely

attractor-sufficient, and an in-silico shift of the corresponding prior should reproduce them. The effect of stimulus intensity, by contrast, should be carried mostly by the biases, as nociceptive drive. In chronic pain the same decomposition can be followed over time, to see whether chronification moves pain from an input-carried to an attractor-level phenomenon, or the reverse (P8).

6. Emergent, biology-like phenomena: hypotheses

Several well-known features of biological learning may follow from the same few properties, above all from self-orthogonalization. Below we highlight a few examples, each a hypothesis to be tested rather than a result. Further-reaching possibilities are collected in Section 11.

- **Primacy effect** [\(HYPOTHESIS\)](#). Because new information is stored only as a residual relative to existing attractors, early material serves as the reference frame for what follows and should be remembered better. This would mirror the primacy effect in human memory [\[Murdock, 1962\]](#), in contrast to classical palimpsest networks, which forget their oldest memories first [\[Nadal et al., 1986, Parisi, 1986\]](#).
- **Extinction, renewal and reinstatement** [\(HYPOTHESIS\)](#). Self-orthogonalization restricts how representations can change:
 - *Acquisition.* A newly learned representation is stored in an attractor approximately orthogonal to all previous ones, so only its residual is stored, and old representations shift as well (remapping) [\[Colgin et al., 2008\]](#).
 - *Context.* Overlap with old memories, for example an association with fear or pain, induces correlations that must be balanced by non-overlapping components to keep the attractors orthogonal. This is how context A becomes part of the memory.
 - *Extinction.* The old memory could only be erased if everything else stayed unchanged, which is practically impossible. Instead a new attractor forms, again orthogonal to the others. Because the stimulus is similar or identical but the outcome differs, self-orthogonalization puts strong pressure on using context to tell the two situations apart: context B.
 - *Renewal and reinstatement.* The old context, or the unconditioned stimulus itself, therefore brings back the original response. Through sequence learning, the network may even anticipate it.
 - *Related work.* This mechanistic account parallels the evidence that extinction is new, context-dependent learning [\[Rescorla and Heth, 1975, Bouton, 2004, Rescorla, 2004\]](#), latent-cause models in which extinction creates a new latent cause [\[Gershman et al., 2010, 2017\]](#), state-splitting accounts of renewal [\[Redish et al., 2007\]](#) and the neurobiology of fear extinction [\[Milad and Quirk, 2012\]](#).
- **Trace conditioning and working memory** [\(HYPOTHESIS\)](#). Self-sustaining attractor activity that persists after the stimulus has gone would bridge the interval in trace conditioning [\[Clark and Squire, 1998\]](#) and support working memory, as in classical persistent-activity models [\[Goldman-Rakic, 1995, Amit, 1997, Wang, 2001\]](#). Activity-silent accounts [\[Stokes, 2015\]](#) provide the contrast to test against.
- **Oscillations during acquisition** [\(HYPOTHESIS\)](#). When a new stimulus is not well explained by any existing attractor, ghost attractor or superposition, only oscillatory sampling across attractors remains. This is slow and costly, because the boundaries between attractors have high free energy, so learning will try to store the new information in an attractor, or at least a ghost, while keeping orthogonality (remapping). The oscillations should then fade as learning proceeds. This may relate to theta-band activity during acquisition and encoding [\[Buzsáki, 2002, Hasselmo et al., 2002, Seidenbecher et al., 2003, Likhtik et al., 2013\]](#), and to the oscillations that emerge in circuits optimized for sampling-based inference [\[Aitchison and Lengyel, 2016, Echeveste et al., 2020\]](#).
- **Curiosity and exploration** [\(HYPOTHESIS\)](#) open. How far the usual active-inference arguments for epistemic value [\[Friston et al., 2015, 2017\]](#) carry over to these networks is unclear, but uncertainty here naturally encourages exploration [\[Oudeyer, 2007, Schmidhuber, 2010, Gottlieb and Oudeyer, 2018\]](#). In this picture a “dark room” [\[Friston and Ao, 2012\]](#) is not predictable: it is a lack of evidence, which leaves spontaneous activity to sample the priors and may produce hallucination-like states. This links to accounts of hallucinations as strong priors [\[Powers et al., 2017, Corlett et al., 2019\]](#) and to pain without input, as in phantom-limb pain [\[Flor et al., 2006\]](#).

- **Reinforcement learning** [\(HYPOTHESIS\)](#). The networks are compatible with reinforcement learning in the way active-inference accounts propose [[Friston et al., 2009](#), [Sutton and Barto, 2018](#)], but this relies on fixed priors set by evolution, such as pain. Pain research is therefore a good test bed [[Büchel et al., 2014](#), [Wager and Atlas, 2015](#), [Tabor et al., 2017](#)].

7. Predictions and falsification criteria

Every prediction below is stated in advance, with the test bed and the result that would count against it. Self-orthogonalization is the system’s most prominent signature, but it is hard to test in high dimensions: random vectors also tend towards orthogonality. Real inputs are almost always structured and correlated, so orthogonality of learned attractors is not trivial. Still, the decisive test we propose is the projector-network signature (P2), not orthogonality alone. All geometric tests use strong null models: phase-randomized time series, symmetry-preserving permutations, covariance-matched Gaussian models, and condition-label shuffling. Analysis plans will be preregistered where possible. For the neuroscience predictions the criterion is mechanistic: counterfactual simulations specified in advance, and the sufficiency test of Section 5.2, rather than predictive accuracy beyond that of existing data-driven models.

#	Prediction	Test bed	Would count against it
P1	Learned attractors are approximately orthogonal, even for correlated inputs	Rest and task fMRI; simulations	Attractor angles indistinguishable from matched null models
P2	Decisive test of self-orthogonalization: the networks approach a Kanter-Sompolinsky projector network, whose attractors coincide with the leading eigenvectors of the coupling matrix	fMRI-fitted networks; simulations	Eigenvector-attractor alignment no better than null models
P3	Orthogonalization depends on the timing of the learning rule: pairing the error with the renewed state orthogonalizes; pairing it with the preceding state merges attractors	Simulations; physical hardware	Orthogonalization with reversed pairing, or none with the correct pairing
P4	Noise encodes precision: a unit's variance equals its gain, and variability tracks inferred precision	Physical hardware; neural and BOLD variability vs responsiveness	No relation between variability and gain or precision
P5	Extinction creates a new, approximately orthogonal attractor rather than erasing the old one; renewal is a context-driven return to it	Threat-learning and extinction fMRI; benchmark battery	Extinction as weakening of the original attractor; renewal not predicted by the fitted landscape
P6	Expectations act as attractor priors: placebo and nocebo shift the posterior towards expectation-congruent states, scaling with prior precision	Placebo and nocebo fMRI	An in-silico shift of the prior does not reproduce the observed placebo and nocebo effects, or these effects are not attractor-sufficient (P21)
P7	Task activity decomposes into network-level (couplings) and input-driven (biases) components	Task fMRI; 7T laminar fMRI (feedback vs feed-forward layers)	Decomposition does not generalize across tasks or maps onto the wrong layers
P8	Chronic pain reflects a deepened, harder-to-leave pain attractor (macro-scale) and/or region-intrinsic change (meso-scale), distinguishable within subjects over time by the attractor-input decomposition (P21)	Longitudinal phantom-limb pain cohort	Neither component tracks chronification
P9	Self-solenoidization: sequential experience produces asymmetric couplings whose flow is approximately solenoidal (along free-energy isocontours) ²⁰ and replays sequences in their learned direction, while attractor occupancies stay those	Sequence-learning fMRI (Helmholtz decomposition of fitted flows); simulations	Transition flow mainly along the free-energy gradient, no directed replay, or occupancies shifted by the asymmetric couplings

Negative results are informative by design: they delimit where the construction applies and point to mechanisms missing from the theory, and they will be published.

8. Technical plan

The three lines share one model class, one set of definitions and one set of predictions. Results from each line are checked against the others, and are often published together.

8.1 Line A – Approach and theory

- **A1. Complete and test the unit-level construction.** Publish the derivation; test its contingent predictions in finite jump-process simulations in which no inference or learning rule is coded: the form of the nonlinearity under different resting laws, fluctuation–gain equality, and the dependence of orthogonalization on learning-rule timing (P3, P4).
- **A2. Composition and hierarchy.** Formalize how evidence at one level acts as the effective field at the next, and how nested timescales give rise to hierarchical message passing. Derive principled coarse-graining criteria: bounds on how well attractors are preserved as a function of within-parcel heterogeneity, and conditions under which the coarse transition law, not only the attractor set, is preserved (Section 4.2). Show how attractors at one level supply the slow timescale for the next, and characterize when the resulting levels are informationally and causally closed (Section 5.1; P17).
- **A3. Action and planning.** Develop the network-level account of action: solenoidal flow from antisymmetric couplings as intrinsic action, and effector-driving nodes as extrinsic action. Study planning as sequence prediction over the learned landscape, and the link between learnable surprise and exploration.
- **A4. Thermodynamic cost.** Relate the quantities of the construction (alignment gap, learnable versus unlearnable surprise) to the heat dissipated during perception and learning, as a bridge to physical implementation. Because each synapse is the thermodynamic Legendre conjugate of a correlation, learning has a thermodynamic price that can be computed in theory and measured in hardware.

Methods: non-parametric information geometry, stochastic thermodynamics, averaging and adiabatic theory, and large-scale simulation.

8.2 Line B – Brains: generative models of large-scale brain dynamics

- **B1. Model fitting.** With the open-source package *attractome*, fit networks to resting-state, task and naturalistic fMRI by score matching with haemodynamic deconvolution, estimating couplings (network-level dynamics) and biases (upstream, mesoscale input). Validate generatively, by simulating held-out data, and against null models. This includes individual-level models from dense, repeatedly sampled single-subject datasets. Before interpretation, the fitted landscapes are tested across resolutions: project and match across nested atlases, measure homogeneity, and compare with random parcellations, in population and single-subject data (Section 4.2; P18–P20).
- **B2. Tests of the theory.** Attractor orthogonality and eigenmode alignment (P1, P2), the couplings–input decomposition (P7), self-solenoidization, transition learning and directed attractor-to-attractor flows in sequence tasks (P9, P13, P22), prior mixing and oscillatory alternation between attractors (P10, P15), spontaneous sampling of priors at rest (P16), and the closure of the attractor-level description (P17) and its corollary, attractor sufficiency (P21). Ultra-high-field (7T) laminar fMRI tests whether inferred priors and inputs map onto feedback and feedforward cortical layers.
- **B3. Learning without forgetting: extinction and renewal (P5).** Fit landscapes before and after acquisition, extinction and renewal in threat-learning paradigms, and test whether extinction adds a new, approximately orthogonal attractor.
- **B4. Expectations as priors (P6).** Model placebo and nocebo effects on pain as precision-weighted combinations of attractor priors and nociceptive input, using placebo-imaging consortium data. Test whether in-silico shifts of the prior, specified in advance, reproduce the observed effects, and whether placebo responses are attractor-sufficient while the effects of stimulus intensity are carried by the input (Section 5.2).

- **B5. Plasticity in pain chronification (P8).** Separate meso-scale (region-intrinsic) from macro-scale (network) plasticity in chronic low back pain, fibromyalgia and longitudinal phantom-limb pain cohorts, using the attractor–input decomposition to follow, within subjects, the level at which pain is carried as it becomes chronic (P8, P21).
- **B6. From predictive to mechanistic neuromarkers.** Brain-based predictive models of pain, cognition and clinical outcomes have matured: they can be replicable with moderate sample sizes [Spisak et al., 2023], and some generalize to independent patient populations [Spisak et al., 2020]. Most, however, rest on data-agnostic machine learning, so their behaviour out of context is hard to predict, their features are hard to map onto mechanisms, and they cannot simulate what has not been measured. The aim here is not to outperform them but to explain what they capture. The sufficiency test (Section 5.2) asks whether the information a predictive model exploits is carried by large-scale attractors or by upstream input, with the predictive model itself as the reference to be matched. Compact model-based features (attractor depth, dwell times, transition costs) are validated externally across cohorts, and counterfactual simulation generates intervention hypotheses that are specified in advance. Real-time fMRI with reinforcement learning from brain feedback provides closed-loop, causal tests of these hypotheses.

Data: large open datasets and cohorts from established collaborations (placebo and pain consortia, two DFG collaborative research centres, an ERA-NET pain consortium).

8.3 Line C – Computers: emergent learning as an alternative route to AI

- **C1. Software and scaling.** Efficient, open implementations; map the networks’ operational regimes (temperature, noise, input precision) and scaling behaviour.
- **C2. A neuroscience-inspired benchmark battery.** Continual learning, extinction, reinstatement and renewal, interference, primacy, reversal and sequence tasks, matched to the paradigms of Line B so that brain and machine are compared on the same tasks (P5, P11, P14), and long-horizon tasks scored against a repeated-sampling baseline (P23). Baselines: standard continual-learning methods (regularization- and replay-based) and backpropagation-trained recurrent networks.
- **C3. Uncertainty and generative inference.** Bayesian reconstruction and denoising as a function of prior and input precision; calibration of the networks’ uncertainty; generalization through prior mixing (P10).
- **C4. Native physical implementation (P12).** Stochastic binary units are the tanh case of the minimal unit, which makes probabilistic-bit and related hardware a natural fit. Steps: simulate hardware-constrained networks; run inference natively on hardware with the local learning rule applied from the measured states; then co-design slowly adapting couplings native to the substrate. Candidate substrates: thermodynamic, memristive, nanomagnetic and related emerging architectures [Camsari et al., 2019, Coles et al., 2023, Harabi et al., 2022], with partners. Device evaluation has so far relied on emulation metrics that rate how faithfully a device reproduces biological synapses, neurons or networks [Bruno et al., 2026]. As a complement, the construction suggests theory-derived figures of merit: the separation of timescales between state and conductance, a bounded, maximum-entropy response, noise variance that tracks gain, bidirectional local updates with the prescribed timing, support for asymmetric couplings, and, at the network level, attractor orthogonality and resistance to forgetting.
- **C5. Closed-loop agency.** Maze, reversal and detour tasks probe world-model formation, planning and autonomous action, and bridge back to behavioural neuroscience.

9. Risks, limitations and fallbacks

The programme is designed so that each risk, if it materializes, still produces an informative result.

Risk	Consequence	Fallback or what is learned
Assumptions of the construction do not hold for a given substrate	The inferential reading is not guaranteed there	Delimits the scope of the theory; the network models remain usable as phenomenological generative models
Orthogonality in brain data is explained by high-dimensional chance or measurement artefacts fMRI is too slow or too noisy for some signatures	P1 and P2 fail Weak tests in Line B	Strong null models detect this; attention shifts to dynamical predictions (P5–P9) Laminar 7T fMRI, dense single-subject sampling, and simulations with matched noise; electrophysiology via collaborators
Counterfactual simulations fail to predict the effects of real manipulations	The fitted landscapes describe brain dynamics without explaining them	Model variants with fixed couplings or biases locate which component fails; the sufficiency analysis still maps which quantities are carried at the attractor level; results are published either way
Networks do not scale on conventional hardware	Line C limited to small and medium problems	Focus on properties (continual learning, calibrated uncertainty) rather than scale; emerging hardware as the long-run route
Emerging hardware cannot realize slow adaptive couplings	Native learning (C4) not reached	Hardware-in-the-loop learning remains publishable; results inform hardware development
Brain and machine share no common signatures	The common-principle hypothesis fails	A clear negative result: the theory may describe a viable route to learning that brains do not use
Attractor-level descriptions of brain data are not closed (P17 fails)	Attractors may be better treated as descriptions than as a level in their own right	The fitted networks remain generative models of regional dynamics; closure is then tested at finer scales and in simulations, where timescale separation is larger (A2)
Few quantities are attractor-sufficient; most are mixed (P21)	No clean assignment of functions to levels	The attractor share remains a graded, theory-derived measure; a consistently large input share would itself show that large-scale attractors carry less of the computation than proposed
Attractors do not match across resolutions (P18 or P19 fails)	fMRI-fitted networks cannot be read as views of one landscape	Claims are restricted to resolutions validated by held-out generative tests; precision fMRI, 7T data and simulations with known ground truth locate where homogeneity fails

Limitations and open questions

Much of this programme rests on heuristics that still have to be made rigorous. We list the main ones so that readers can weigh them.

- **Sufficient, not necessary.** The unit-level construction shows one route from physics to inference and learning. It does not show that brains, or any particular material, take this route.
- **Strong assumptions.** Timescale separation, bounded coordinates and leading-order interactions are idealizations. Where they fail, the inferential reading is not guaranteed.

- **Heuristics not yet derived.** Self-solenoidization, the duality with antisymmetric couplings, prior mixing through oscillatory alternation, and the role of expected free energy in intrinsic action are hypotheses, not results.
- **Not an exact gradient.** The network learning rule is not the exact gradient of a single objective, so its stationary behaviour is characterized only approximately.
- **Minimal units.** One-dimensional, continuous-Bernoulli units are the simplest case; multidimensional or spiking units need extensions.
- **Measurement.** fMRI is slow, indirect and confounded by vascular and physiological signals. Results depend on coarse-graining choices, since a parcellation shows only distinctions that live across parcels (Section 4.2), and orthogonality in high dimensions needs strong null models.
- **Scale.** Demonstrations so far are small to medium in scale. We make no claim of competitiveness with large deep-learning systems.
- **Speculative links.** The broader view of life as stacked timescales and the connections to quantum-like geometry and to consciousness are collected in the Outlook (Section 11). The rest of the programme does not depend on them.
- **The roots, not the whole tree.** The programme aims to explain where the brain’s capacities come from, not the full complexity with which a particular brain realizes them (Section 1.5). Success would leave most of neuroscience’s hard problems open, while giving them, we hope, a common starting point.
- **Levels and causation.** Whether attractors are causes or descriptions ultimately touches philosophical questions that the programme does not settle [Kim, 1998]. We argue only for causal relevance in senses that can be tested: the creation of timescales, the effects of interventions, and closure (Section 5.1; P17).

10. Relation to other approaches

The programme is meant to complement neighbouring approaches rather than compete with them. The table summarizes where each meets it and what the programme adds.

Approach	Central object	Where it meets this programme	What this programme adds
Neuroconnectionism, task-optimized deep networks	Architecture, objective and training data that make networks brain-like	Shared aim: computational models tested against brain and behavioural data	A first-principles route to the learning rule and dynamics; whole-brain generative modelling; continual learning by construction
Biophysical whole-brain models (neural mass, neural field)	Biophysics at one chosen scale	Shared target: large-scale brain dynamics	A scale-agnostic family of rules that complements scale-specific mechanisms; task fidelity, as networks trainable as learners
Data-driven reconstruction of brain dynamics (recurrent networks fitted to recordings)	Generative dynamical models learned from data	Shared goal: generative models of brain dynamics	A model class derived from first principles, with interpretable priors, inputs and learning
Free Energy Principle, active inference	Variational free energy minimization by self-organizing systems	The networks minimize variational free energy at two timescales	Sufficient physical conditions for one minimal system; no Markov blanket required; falsifiable predictions
Bayesian brain, predictive coding	Posterior inference; prediction errors	Prediction errors, posteriors and precisions appear in the construction	A physical origin for prediction errors and precision; the timing of the learning rule
Sampling-based probabilistic computation	Neural variability as samples from a posterior	Fluctuations sample the posterior at two timescales	Variance equals gain; noise as precision; a learned Fisher metric
Hopfield networks, Boltzmann machines, dense associative memories	Energy landscapes; contrastive learning; storage capacity	The networks generalize stochastic Hopfield networks and Boltzmann machines	Local, single-phase learning; asymmetric couplings; self-orthogonalization
Landscape-flux theory of non-equilibrium systems	Decomposition into potential and curl flux	Symmetric and antisymmetric couplings split gradient and solenoidal flow	A learning rule expected to produce this split on-line (self-solenoidization)
Latent-cause models of learning and extinction	Inference over hidden causes of experience	Extinction as the formation of a new cause	An attractor-level mechanism with neural predictions
Physical learning (equilibrium propagation, coupled learning)	Learning rules native to physical networks	Shared aim: learning native to physics	Unsupervised, generative learning derived from first principles; a link to brain data
Continual-learning methods in machine learning	Avoiding catastrophic forgetting	Shared benchmarks	Resistance to forgetting as an emergent property, not an add-on
Information-theoretic accounts of emergence (causal emergence; informational and causal closure)	Macroscopic variables that are self-contained, or causally more informative than the micro level	Attractors as candidate closed macro-variables of brain dynamics	A mechanism that generates such levels from within, through learned attractors with their own timescales; an fMRI test (P17)
Connectome-based attractor and energy-landscape models of brain data	Energy landscapes or attractors of Ising- or Hopfield-type models fitted to, or built from, brain connectivity	The same model family and the same data [Watanabe et al., 2013, Ezaki et al., 2017, Gu et al., 2018, Fasoli et al., 2026]	A derived learning rule; attractors read as Bayesian priors; self-orthogonalization as a testable signature (P1, P2)
Evolution and development as learning	Associative learning by networks of interacting components over evolutionary and developmental time	Learning as a property of organized matter, across levels [Watson et al., 2011, Vanchurin et al., 2022, Watson and Levin, 2023, Fields and	A minimal physical construction of the elementary step, with tests in brains and machines; a reading of evolution as refining, rather than in-

Eight criteria summarize what the construction is meant to achieve. They are listed, with the sections where each is addressed, in *Desiderata: what the construction should achieve*.

Desiderata: what the construction should achieve			
#	Criterion	What it requires	Addressed in
D1	Physically grounded	Inference and learning follow from stated physical assumptions about coarse observables, not from algorithms imposed from outside.	Section 3
D2	Scale-agnostic	Rules of the same family apply, approximately, at any level of description (unit, region, network), so that the model class transfers across scales and coarse measurements such as fMRI are legitimate targets. Ideally, the model also explains how new levels of description arise from within the system.	Sections 4.2, 5
D3	Local	Learning uses only quantities available at the unit: no global loss and no error transport.	Sections 3.3, 4.1
D4	Constructive and generative	The model can be built from its stated rules, as a simulation or in hardware, ideally running natively on physical substrates with noise as a resource. As a brain model, it is not only fitted to data but can also generate data that were not measured, including the effects of counterfactual interventions.	Sections 5.2, 8
D5	Trainable	The same model class learns tasks: sequences, continual learning, inference under uncertainty.	Sections 2, 4.3, 8.3
D6	Probabilistic	Uncertainty is represented natively, at both fast (states) and slow (structure) timescales.	Sections 3.3, 4.4
D7	Falsifiable	Assumptions are explicit, each can fail, and the model makes quantitative predictions with stated failure criteria.	Sections 3.2, 7, 9
D8	Interpretable	Model components map onto computational roles (priors, likelihoods, prediction errors) and onto measurable quantities.	Sections 3.3, 4, 5.2

Intellectual lineage and acknowledgements

This programme stands on many traditions, and most of its ideas are new combinations of theirs rather than independent innovations. We acknowledge in particular:

- **Statistical physics and information geometry:** maximum entropy [Jaynes, 1957], fluctuation–dissipation [Kubo, 1966], projection operators and memory kernels [Zwanzig, 1961, Mori, 1965], averaging and multiscale methods [200, 2008], synergetics and slaving [Haken, 1983], centre manifolds [Carr, 1981], non-parametric information geometry [Pistone and Sempi, 1995] and information geometry more broadly [Amari, 1998, 2016].
- **The free energy principle and active inference:** Friston [2010, 2019], Palacios et al. [2020], Parr et al. [2022], Friston et al. [2023a,b], Ramstead et al. [2023], Sakthivadivel [2022] and Friston et al. [2025]; and the critical literature that sharpened our formulation [Andrews, 2021, Biehl et al., 2021, Millidge et al., 2021, Aguilera et al., 2022, Bruineberg et al., 2021].
- **The Bayesian brain and predictive coding:** Rao and Ballard [1999], Knill and Pouget [2004], Friston [2005], Feldman and Friston [2010], Bastos et al. [2012], Whittington and Bogacz [2017], Song et al. [2024] and Senn et al. [2024]; and hierarchical Bayesian models of learning under uncertainty [Mathys, 2011].
- **Sampling-based probabilistic computation:** Hoyer and Hyvärinen [2003], Fiser et al. [2010], Berkes et al. [2011], Buesing et al. [2011], Kappel et al. [2015], Aitchison and Lengyel [2016], Orbán et al. [2016], Echeveste et al. [2020] and Hénaff et al. [2020]; and probabilistic population codes as an alternative [Ma et al., 2006].
- **Attractor networks and energy-based models:** Hopfield [1982], Amit et al. [1985], Ackley et al. [1985], Smolensky [1986], Personnaz et al. [1986], Sompolinsky and Kanter [1986], Kleinfeld [1986], Kanter and Sompolinsky [1987], Amit [1989], Hinton [2002], Barra et al. [2012], Krotov and Hopfield [2016], Ramsauer et al. [2020], Khona and Fiete [2022], Hoover et al. [2025] and Clark [2026]; Hebbian unlearning and dreaming [Hopfield et al., 1983, Fachechi et al., 2019, Serricchio et al., 2025]; and perturbation evidence for attractor dynamics in brains [Kim et al., 2017, Inagaki et al., 2019, Finkelstein et al., 2021].
- **Local Hebbian and anti-Hebbian learning:** Oja [1982], Földiák [1990], Bi and Poo [1998], Gerstner et al. [2018], Pehlevan and Chklovskii [2019] and Lillicrap et al. [2020].
- **Non-equilibrium dynamics and metastability:** potential–flux decompositions [Ao, 2004, Kwon et al., 2005, Wang et al., 2008, Friston and Ao, 2012, Yan et al., 2013], irreversible samplers [Ma et al., 2015, Rey-Bellet and Spiliopoulos, 2015], metastability and itinerancy [Kelso, 1995, Tsuda, 2001, Rabinovich et al., 2008, Tognoli and Kelso, 2014, Hancock et al., 2024] and broken detailed balance in the brain [Lynn et al., 2021, Kringelbach et al., 2024] and the non-equilibrium thermodynamics of asymmetric networks [Aguilera et al., 2023].
- **Large-scale brain dynamics:** resting-state networks and their relation to tasks [Raichle et al., 2001, Fox et al., 2005, Smith et al., 2009, Cole et al., 2016, Margulies et al., 2016, Dosenbach et al., 2025]; whole-brain models [Deco et al., 2008, 2010, Honey et al., 2009, Sanz Leon et al., 2013, Breakspear, 2017]; maximum-entropy and energy-landscape models [Schneidman et al., 2006, Watanabe et al., 2013, Ezaki et al., 2017, Gu et al., 2018]; connectome-based attractor models of fMRI [Englert et al., 2026, Fasoli et al., 2026]; attractor manifolds and timescale separation [Runfola et al., 2025, Athanasiadis et al., 2025]; metastable attractor dynamics and expectation [Mazzucato et al., 2019]; parcellation and its evaluation [Gordon et al., 2014, Eickhoff et al., 2018, Urchs et al., 2019] and precision functional mapping [Gordon et al., 2017]; brain-state models [Vidaurre et al., 2017]; data-driven reconstruction of brain dynamics [Koppe et al., 2019, Singh et al., 2020, Durstewitz et al., 2023]; and the debate between bottom-up and top-down neuroscience [Rosas et al., 2025].
- **Continual learning:** McCloskey and Cohen [1989], McClelland et al. [1995], French [1999], Kirkpatrick et al. [2017], Shin et al. [2017], Zeng et al. [2019], Hadsell et al. [2020], van de Ven et al. [2020], Libby and Buschman [2021], Flesch et al. [2022] and Chandra et al. [2025].

- **Learning, extinction and expectation:** Rescorla and Heth [1975], Bouton [2004], Rescorla [2004], Redish et al. [2007], Gershman et al. [2010], Milad and Quirk [2012], Büchel et al. [2014], Wager and Atlas [2015] and Tabor et al. [2017].
- **Physical, neuromorphic and thermodynamic computing:** Strukov et al. [2008], Still et al. [2012], England [2015], Prezioso et al. [2015], Goldt and Seifert [2017], Scellier and Bengio [2017], Camsari et al. [2019], Borders et al. [2019], Conte et al. [2019], Hooker [2021], Stern et al. [2021], Schuman et al. [2022], Coles et al. [2023], Harabi et al. [2022], Stern and Murugan [2023] and Stern et al. [2025].
- **NeuroAI:** Hassabis et al. [2017], Lake et al. [2016], Richards et al. [2019], Ali et al. [2022], Doerig et al. [2023] and Zador et al. [2023].
- **Scale-free and basal cognition:** Lyon [2005], Watson et al. [2011], Levin [2019], Fields and Levin [2022], Vanchurin et al. [2022], Watson and Levin [2023], Friston et al. [2025] and Yampolskaya and Mehta [2026].
- **Emergence, levels and causation:** hierarchy and near-decomposability [Simon, 1962], “more is different” [Anderson, 1972], order parameters and circular causality [Haken, 1983, Kelso, 1995], hierarchies of timescales in the brain [Kiebel et al., 2008, Murray et al., 2014] and in recurrent networks [Yamashita and Tani, 2008], interventionist causation [Woodward, 2004], causal emergence and closure [Hoel et al., 2013, Rosas et al., 2020, 2024]; and, for the opposing view, the causal exclusion argument [Kim, 1998].

We are grateful to the many colleagues whose work and conversations shaped these ideas. Any errors in combining them are our own.

11. Outlook: further-reaching possibilities

The rest of the programme does not depend on the ideas in this section **(SPECULATIVE)**.

We state them because they follow naturally from the construction, and because stating them openly makes it easier to see where it might lead, or where it might break.

Life as stacked timescales. From the point of view of Section 5, life stacks timescales, from molecules and cells to circuits, behaviour, development and evolution, and it appears to do so in a way that largely respects the boundaries between levels (Section 5.1). Hopfield-type attractor dynamics indeed recur across biology, from cell differentiation and epigenetic memory to molecular self-assembly [Yampolskaya and Mehta, 2026]; local adaptation of interactions can produce associative memory at the scale of a whole system [Watson et al., 2011]; and evolution and development have been cast as learning across nested levels and timescales [Vanchurin et al., 2022, Watson and Levin, 2023]. That living systems largely respect these boundaries is a great gift for scientific understanding, and it is what makes the programme’s bet plausible: if the same elementary step recurs at each level, understanding it once gives a handle on many.

Evolution as refining a law rather than inventing rules. If the elementary step is a property of organized matter rather than a biological invention (Sections 1.5 and 3.4), evolution may have worked less like an engineer designing rules and more like a craftsman perfecting a material. It would start from structures that align only weakly with their environment and favour substrates in which the same law operates with more headroom: more reliable separation of timescales, wider ranges of coupling, richer architectures. Because better-aligned systems can in turn build better substrates, the gains could compound, so that alignment improves at an accelerating, perhaps exponential, rate over evolutionary time. This echoes views of evolution and development as one scale-free process acting on materials that are already competent [Fields and Levin, 2020, Levin, 2023], and of evolution as learning [Watson and Levin, 2023, Vanchurin et al., 2022]. Whether any of this can be made quantitative is open.

Quantum-like geometry. The rotational, solenoidal part of the dynamics (Section 4.5) leads into the territory of tangent and cotangent bundles. On a dually flat statistical manifold, Dombrowski’s construction gives the tangent bundle a Kähler structure [Dombrowski, 1962, MOLITOR, 2012, Molitor, 2013], the same kind of geometry that underlies the geometric formulation of quantum mechanics [Kibble, 1979, Ashtekar and Schilling, 1999]. Closed solenoidal loops may then accumulate geometric, Berry-like phases [Berry, 1984], which are known to arise in classical stochastic kinetics [Sinitzyn and Nemenman, 2007]. Whether this formal

kinship can account for quantum-like phenomena in cognition and decision making [Busemeyer and Bruza, 2012, Pothos and Busemeyer, 2013] is open.

Consciousness. Attractors offer a plausible latent dimension for a global workspace [Baars, 1988, Dehaene et al., 1998, Mashour et al., 2020, Whyte and Smith, 2021] or for the “inner screen” of Ramstead et al. [2023]. We make no claim here beyond pointing out these links.

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